Unit 3 - Calculus
3.1 Limits and Derivatives

Graham West

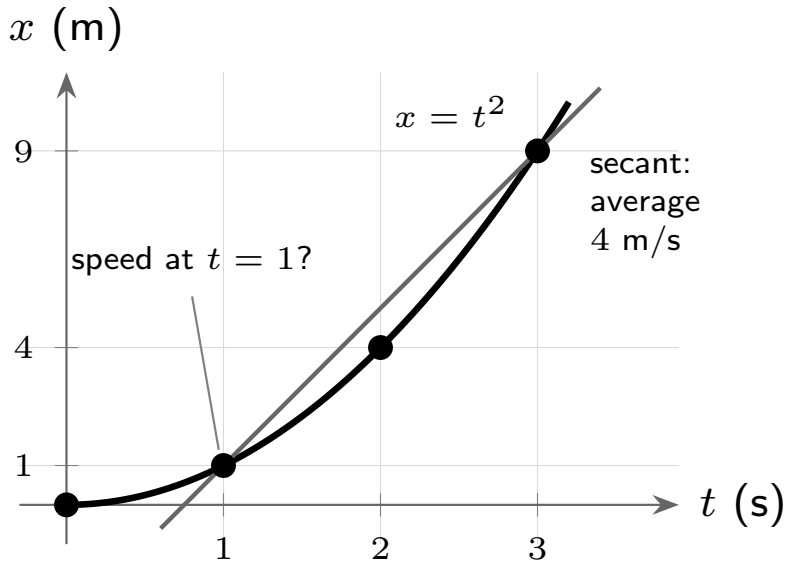
MSDS 520: Math and Stats Foundations for Data Science
Meharry Medical College

Outline

1. Calculus: the Study of Change
2. Average Rate of Change
3. From Secant to Tangent
4. The Derivative
5. Derivative Rules
6. The Chain Rule
7. Partial Derivatives
8. Review

Calculus: the Study of Change

A Problem to Motivate Us



A car pulls away from a stop sign

Position $x(t) = t^2$ m after t s. From $t = 1$ to $t = 3$ it covers 8 m in 2 s: an **average** of 4 m/s.

But a speedometer reads one instant

What does it read at exactly $t = 1$? Speed needs **two** instants; at one, distance and time are both 0:

$$\text{speed} = \frac{0}{0} \quad ???$$

This lecture's question

How do we measure a rate of change at a **single instant**? $\Rightarrow$ the **derivative**

Where This Shows Up in Data Science

Rates of change are everywhere

- **Velocity**: meters per second.
- **Drug elimination**: how fast a drug's blood concentration falls, mg/L per hour.
- **Epidemics**: new cases per day, and whether that number is itself growing.
- **Temperature across a room**: degrees per meter. A rate can be with respect to position, not just time.

Why a data scientist needs it

- Training a model means asking: if I nudge this **weight**, how fast does the **error** change?
- That rate is a derivative. Every model you train, from linear regression to a neural network, is trained by following derivatives downhill (Unit 3-2).

What We Will Build

The plan

1. **Average** rate of change: secant lines and the **difference quotient**.
2. Shrink the interval: **just enough** about limits to get past $0/0$.
3. The **derivative**: its definition, its notation, and six examples worked from the definition.
4. **Rules** (power, sum, product, quotient): shortcuts, so we never have to take the limit by hand again.
5. The **chain rule**, for functions inside functions. Every error function in Unit 3-2 needs it.
6. **Partial derivatives**: for a function of two variables (Unit 1-3), a slope in each direction.

Notebooks for this lecture: `U3-2_Limits-1`, `U3-3_Derivatives-1` and `-2`, `U3-6_Multivariable-1` and `-2`.

Average Rate of Change

Rate of Change Is Rise over Run

Definition: average rate of change

The **average rate of change** of f from x_1 to x_2 is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}.$$

- Δ ("delta") means "change in".
- Units: output units **per** input unit.

The car

$$\frac{x(3) - x(1)}{3 - 1} = \frac{9 - 1}{2} = 4 \text{ m/s}$$

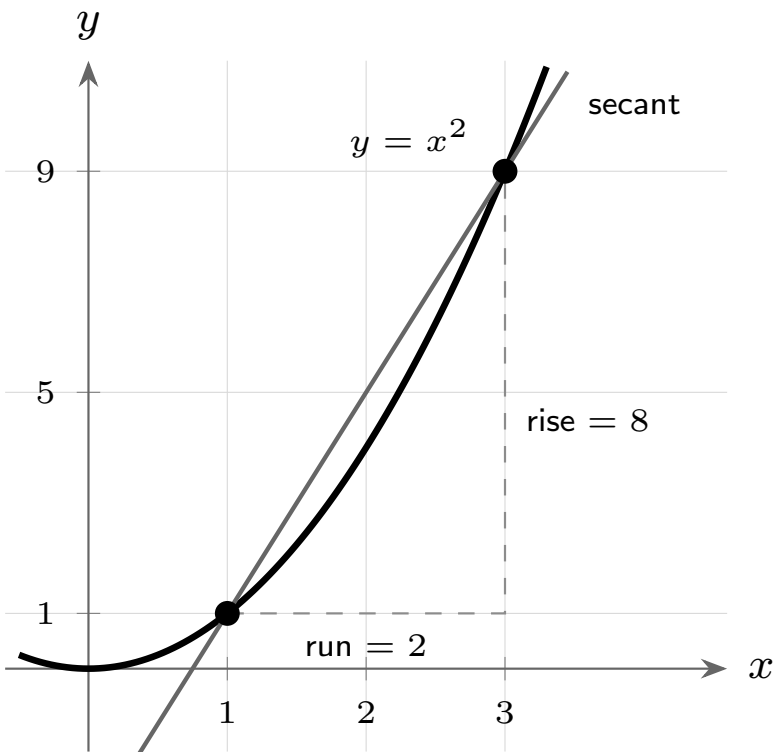
A drug clearing the blood

C falls from 8 to 2 mg/L over 3 hours:

$$\frac{2 - 8}{3} = -2 \text{ mg/L per hour}$$

Negative: the concentration is **falling**.

Secant Lines



Definition: secant line

A **secant line** crosses a graph at **two** points.

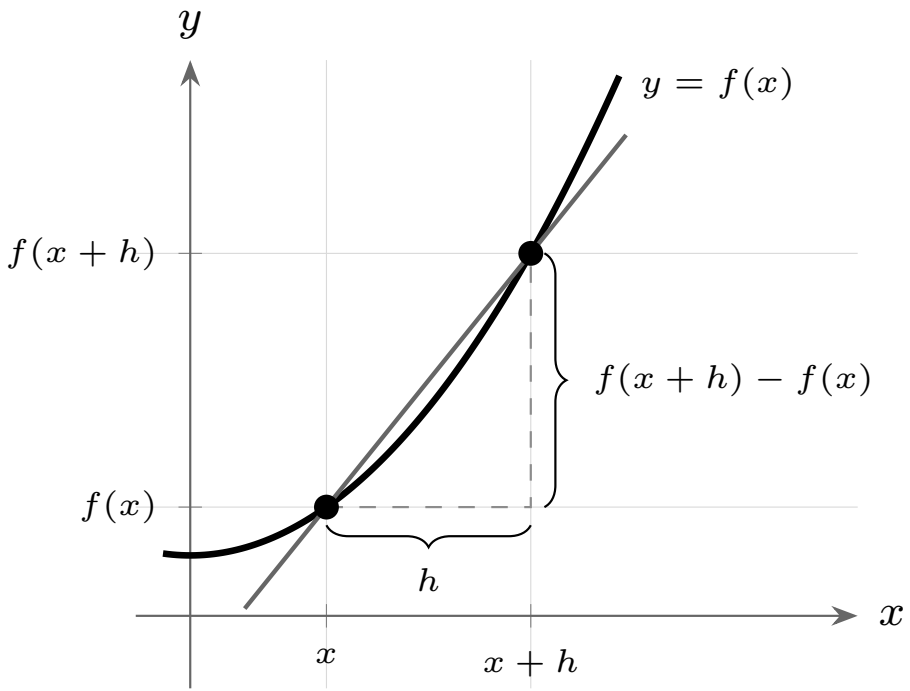
- Its slope is the **average** rate of change between them.
- Here: from $(1, 1)$ to $(3, 9)$,

$$\text{slope} = \frac{8}{2} = 4.$$

The catch

The curve is steeper at $x = 3$ than at $x = 1$. A single average hides that.

The Difference Quotient



Same slope formula, new names: start at x, take a step of size h.

Definition: difference quotient

$$\frac{f(x+h) - f(x)}{h}$$

- h is the **run**: the step.
- $f(x+h) - f(x)$ is the **rise**.
- It is the slope of the secant from x to $x + h$.

Where this is going

Make h smaller and smaller.

Exercise: Average Rates of Change

Try it

1. For $f(x) = x^2$, compute the difference quotient at $x = 1$ with $h = 1$, then with $h = 0.5$. What is happening to the slope?
2. A patient's temperature is $101.3°$F at 8 am and $99.5°$F at 2 pm. Find the average rate of change, with units, and say what it means.
3. For the line $f(x) = 3x + 2$, simplify $\dfrac{f(x+h) - f(x)}{h}$ for **any** h. What do you notice?

Exercise: Solution

Answers

1. With $h = 1$: $\dfrac{f(2) - f(1)}{1} = \dfrac{4 - 1}{1} = 3.$

   With $h = 0.5$: $\dfrac{f(1.5) - f(1)}{0.5} = \dfrac{2.25 - 1}{0.5} = 2.5.$

   As h shrinks, the secant gets **less steep**: it is settling toward the steepness right at $x = 1$.

2. Six hours pass, so

   $$\frac{99.5 - 101.3}{6} = \frac{-1.8}{6} = -0.3 \ ^\circ\text{F per hour.}$$

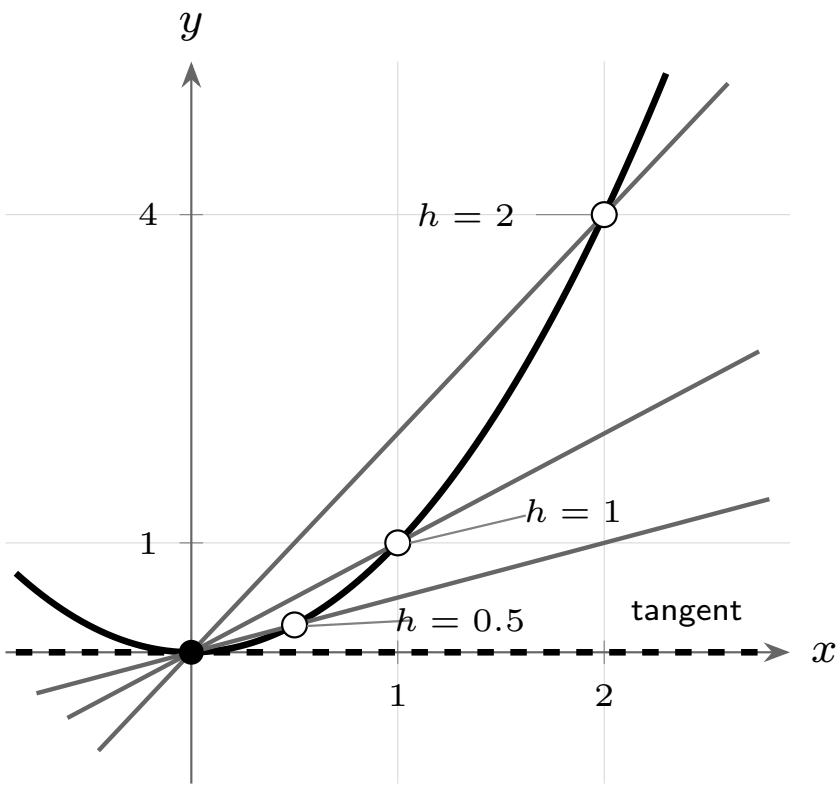
   The fever fell, on average, 0.3 degrees every hour.

3. $\dfrac{3(x+h) + 2 - (3x+2)}{h} = \dfrac{3x + 3h + 2 - 3x - 2}{h} = \dfrac{3h}{h} = 3.$

   The answer is 3 for every h. A secant of a straight line is the line itself, so its slope never changes.

From Secant to Tangent

Shrink the Step



$f(x) = x^2$ at $x = 0$

h	secant slope
2	2
1	1
0.5	0.5
0.1	0.1
0.01	0.01

What we see

- The points labelled by h slide down the curve toward $(0, 0)$.
- The secants flatten onto the dashed tangent, the x-axis: their slopes home in on 0.

Notebook `U3-3_Derivatives-1_SecantToTangent`, §2 and §3

The Problem with $h = 0$

Why not just set $h = 0$?

$$\frac{(0+0)^2 - 0^2}{0} = \frac{0}{0}$$

- Zero rise over zero run: the speedometer problem again.
- $0/0$ is **undefined**. It is not 0, not 1, not anything.

The way around it

- We may **not** set $h = 0$.
- We **may** make h as close to 0 as we like, and watch where the slope is heading.
- "Where it is heading" has a name: a **limit**.

Limits: Just Enough

Definition (informal): limit

$$\lim_{h \to 0} g(h) = L$$

means $g(h)$ gets **as close to** L **as we like** once h is close enough to 0, **without** h **ever equalling** 0. Read it "the limit of $g(h)$ as h goes to 0".

From both sides

h	slope	h	slope
0.1	0.1	-0.1	-0.1
0.01	0.01	-0.01	-0.01
0.001	0.001	-0.001	-0.001

Both sides head to the same place: the limit is 0.

What a limit ignores

- The value **at** $h = 0$ itself.
- That is the whole point: the quotient **has** no value there, yet it still heads somewhere.

Notebook `U3-2_Limits-1_ApproachingALimit`, §1 and §3

Computing a Limit: Simplify, Then Shrink

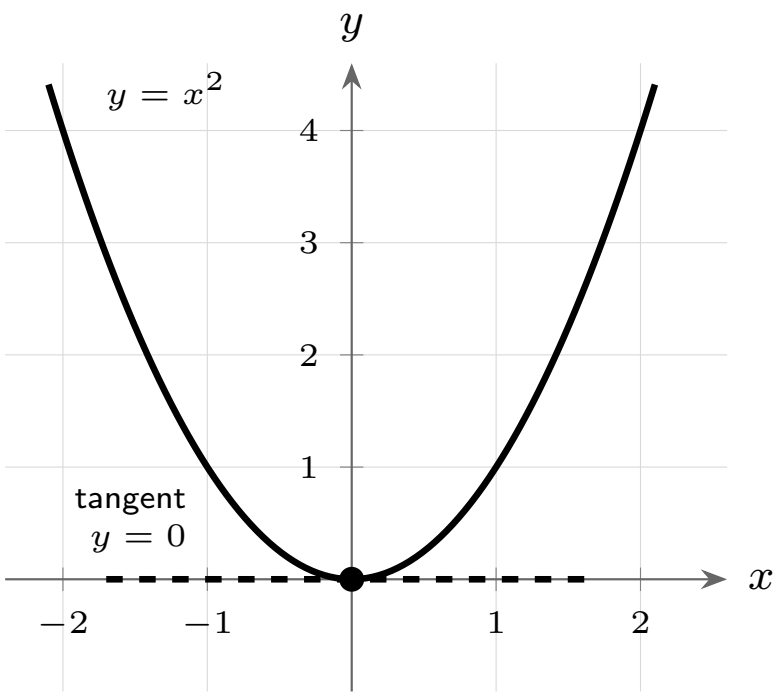
Worked: the slope of x^2 at $x = 0$

$$\begin{aligned}
\frac{(0+h)^2 - 0^2}{h} &= \frac{h^2}{h} && \text{expand} \\
&= \frac{h \cdot h}{h} && \text{factor out } h \\
&= h && \text{cancel, allowed since } h \neq 0 \\
\lim_{h \to 0} h &= 0 && \text{now let } h \text{ shrink}
\end{aligned}$$

The move

- The h that caused $0/0$ **cancels**. After that, nothing stops h from going to 0.
- Every limit in this unit goes this way: simplify first, then let h shrink.
- The same steps work at any point. The next section runs them for every x at once.

The Tangent Line



Definition: tangent line

The line the secants approach as $h \to 0$.

- Touches the graph at **one** point.
- Runs in the **same direction** as the curve.
- Slope $=$ the **instantaneous** rate of change.

Back to the car

At $t = 0$ the tangent is flat: the speedometer reads **0 m/s**, the car at rest as it pulls away. The $t = 1$ reading comes in the next section.

Notebook `U3-3_Derivatives-1_SecantToTangent`, §4

The Derivative

The Derivative

Definition: derivative

The **derivative** of f at x is the limit of the difference quotient:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

- The slope of the **tangent line** at x.
- The **instantaneous rate of change** of f at x, in output units **per** input unit: for a drug level over time, mg/L per hour.
- To **differentiate** is to find the derivative. If the limit exists, f is **differentiable** at x.

Secant versus tangent

	secant	**tangent**
points on the graph	two	one
step	$h \neq 0$	$h \to 0$
slope means	average rate	instantaneous rate

Notation

Four ways to write the same derivative

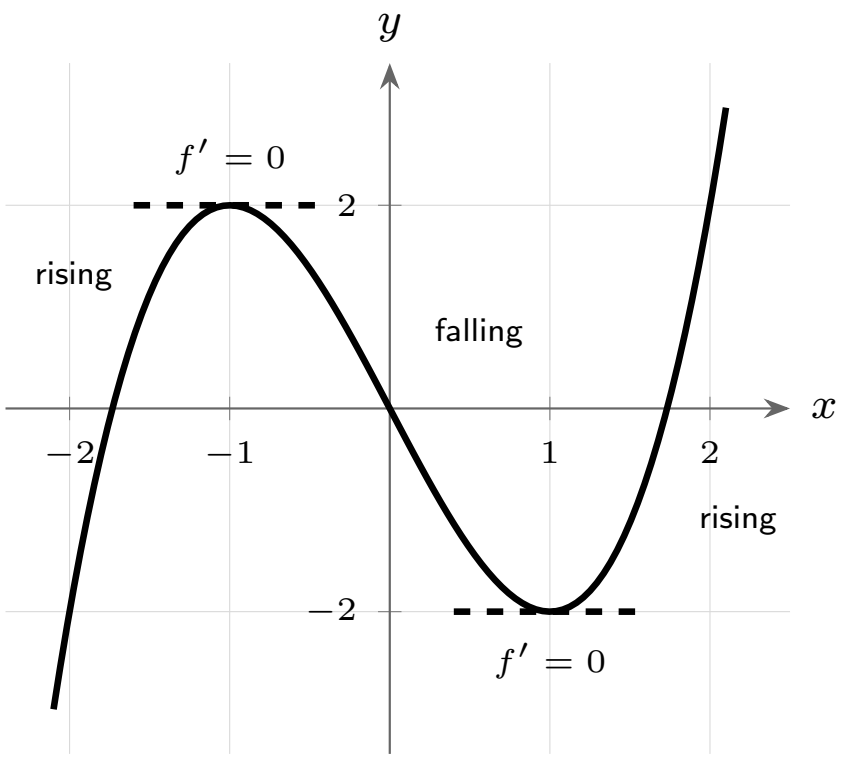
$f'(x)$	"f prime of x"
$\dfrac{df}{dx}$	a tiny rise df over a tiny run dx: the limit of $\Delta f/\Delta x$
$\dfrac{d}{dx}f(x)$	an operation applied to f: in goes a function, out comes a function
$\dot{x}(t)$	"x dot": only for rates **with respect to time**, $\dot{x} = \dfrac{dx}{dt}$

If $y = f(x)$, you will also see y' and $\dfrac{dy}{dx}$. The car's velocity is $\dot{x}(t)$.

Several inputs: partial derivatives

$\dfrac{\partial f}{\partial x}$, "partial f, partial x". The curly ∂ says f has **other** inputs, held fixed while only x moves. The last section of this lecture builds this.

Reading a Derivative off a Graph



The sign of f'

$f'(x) > 0$ graph **rising**

$f'(x) < 0$ graph **falling**

$f'(x) = 0$ graph **flat**

Why we care

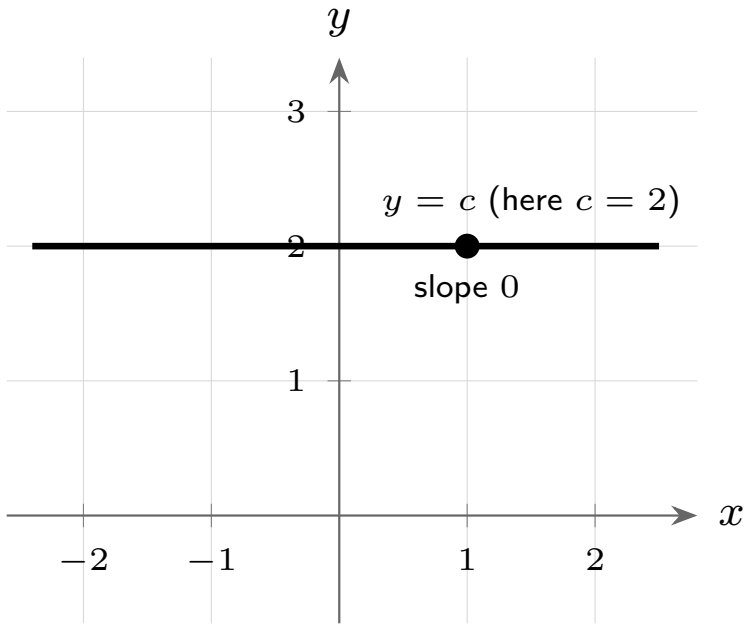
The flat spots are the peaks and valleys. Finding the lowest valley is how models are fit: Unit 3-2 hunts for these.

Example: $f(x) = c$

From the definition

$f(x+h) = c$ too: there is no x to replace.

$$f'(x) = \lim_{h \to 0} \frac{c - c}{h}$$
$$= \lim_{h \to 0} \frac{0}{h} = 0$$



What it says

A constant never changes, so its rate of change is 0 everywhere. The graph is flat, and the tangent at every point is the line itself.

Watch out

$\frac{0}{h} = 0$ because $h \neq 0$. The limit never divides by zero.

Example: $f(x) = x$

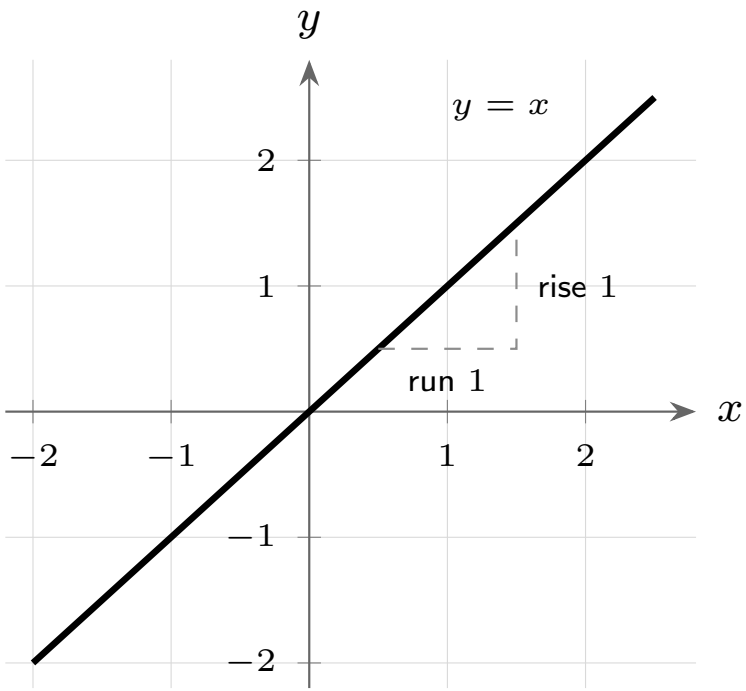
From the definition

$f(x+h) = x + h.$

$$f'(x) = \lim_{h \to 0} \frac{(x+h) - x}{h}$$
$$= \lim_{h \to 0} \frac{h}{h} = 1$$

What it says

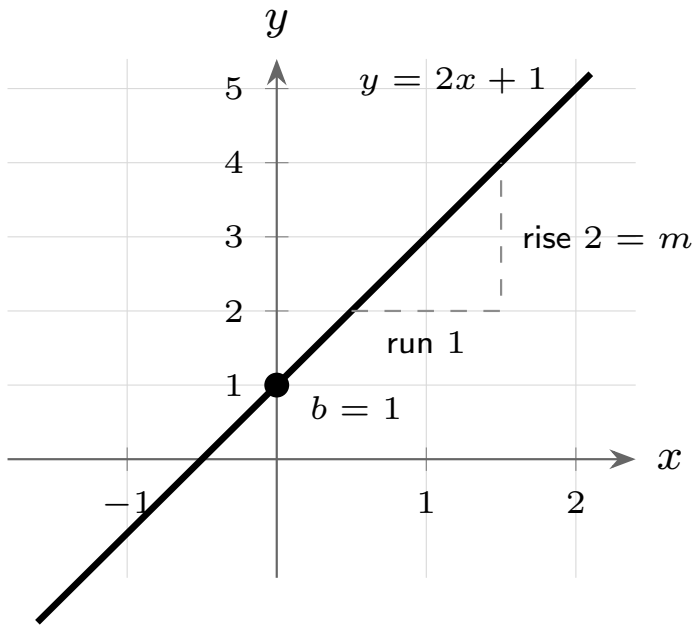
The line $y = x$ has slope 1 everywhere: up one for every one across.



Example: $f(x) = mx + b$

From the definition

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{m(x+h) + b - (mx + b)}{h} \\
&= \lim_{h \to 0} \frac{mx + mh + b - mx - b}{h} \\
&= \lim_{h \to 0} \frac{mh}{h} = m
\end{aligned}
$$



What it says

- A line's derivative is its **slope** m, the same at every x. The intercept b slides the line up or down without tilting it, so it drops out.
- c and x were the special cases $m = 0$, and $m = 1$ with $b = 0$.

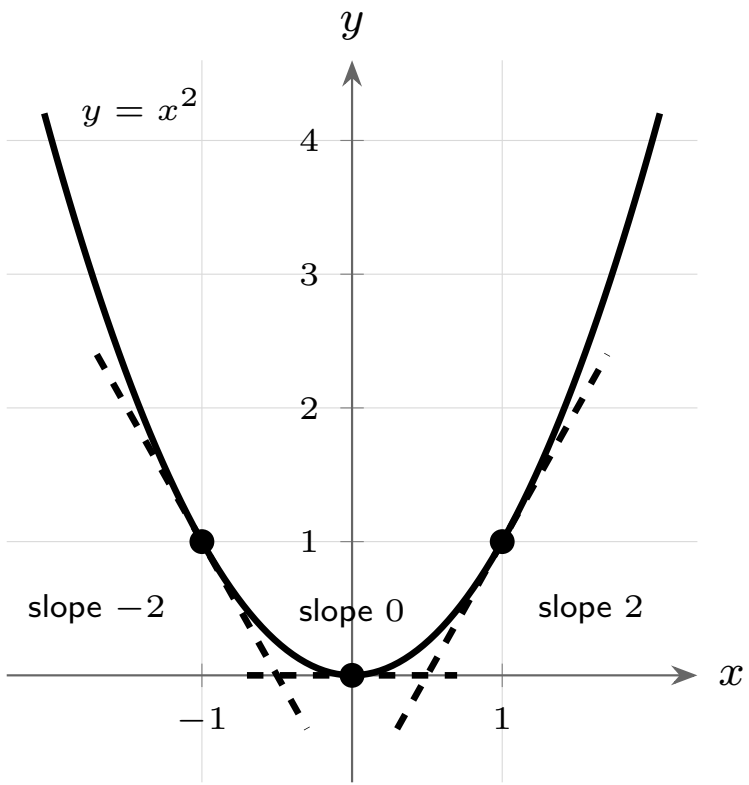
Example: $f(x) = x^2$

From the definition

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{h\,(2x+h)}{h} \\
&= \lim_{h \to 0} (2x + h) = 2x
\end{aligned}
$$

The derivative is a new function

$f'(x) = 2x$ gives the slope at **every** x at once: 0 at $x = 0$, then $2, 4, 6$ at $x = 1, 2, 3$. The car's speedometer at $t = 1$: **2 m/s**.



Example: $f(x) = \sin x$

From the definition

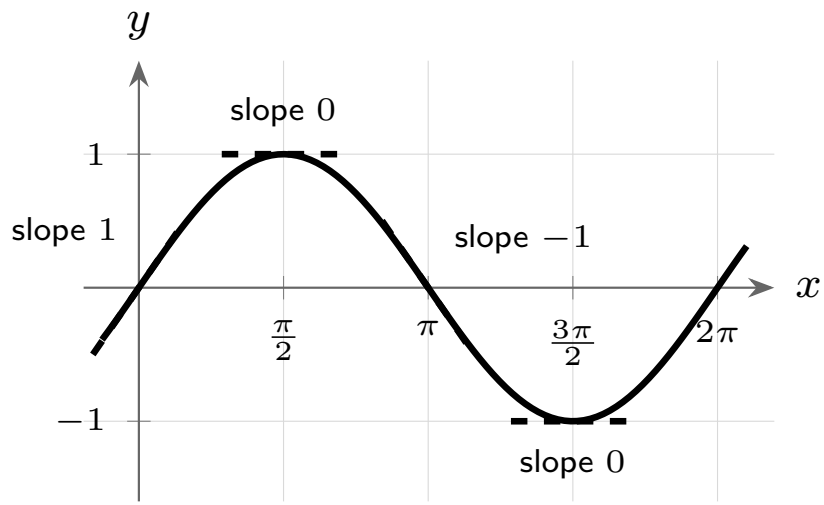
The angle-addition identity, $\sin(x+h) = \sin x \cos h + \cos x \sin h$, splits the quotient:

$$\frac{\sin(x+h) - \sin x}{h} = \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}$$

Two limits, read off a table:

h	$\frac{\sin h}{h}$	$\frac{\cos h - 1}{h}$
0.1	0.9983	-0.0500
0.01	0.99998	-0.0050
$\to 0$	$\to 1$	$\to 0$

So $f'(x) = \sin x \cdot 0 + \cos x \cdot 1 = \cos x$.



What it says

The slopes $1, 0, -1, 0$ are exactly the values of $\cos x$ at $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$. (This needs x in radians.)

Example: $f(x) = 1/x$

From the definition

Combine the fractions in the numerator first:

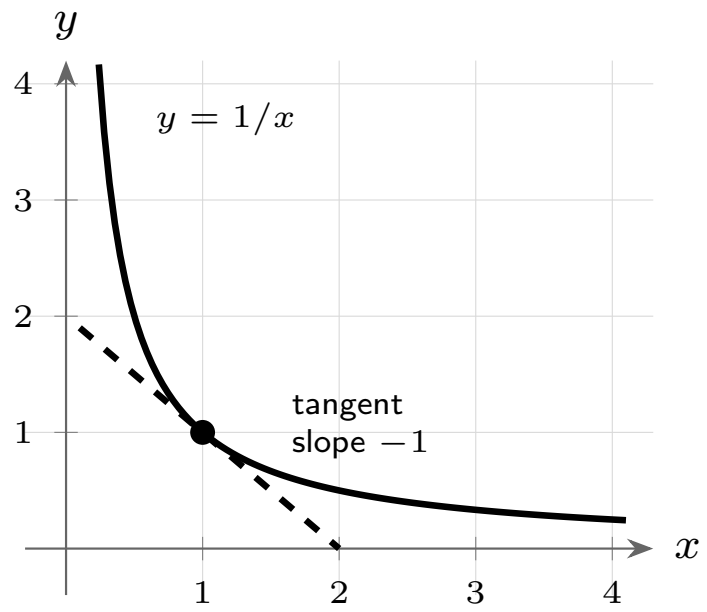
$$\frac{1}{x+h} - \frac{1}{x} = \frac{x-(x+h)}{x\,(x+h)} = \frac{-h}{x\,(x+h)}$$

Now divide by h, which cancels:

$$\frac{f(x+h)-f(x)}{h} = \frac{-1}{x\,(x+h)}$$

Let $h \to 0$:

$$f'(x) = -\frac{1}{x^2}$$



What it says

- $-1/x^2 < 0$: $1/x$ is **always falling**.
- Steep near 0, nearly flat far out.

Exercise: From the Definition

Try it

1. Use the limit definition to find the derivative of $f(x) = 3x^2$.
2. Use the limit definition to find the derivative of $f(x) = x^2 + x$.
3. A drug's blood concentration has $\dfrac{dC}{dt} = -1.5$ mg/L per hour at $t = 4$ hours. In words, what does that mean?

Exercise: Solution

Answers

1. $$\frac{3(x+h)^2 - 3x^2}{h} = \frac{3x^2 + 6xh + 3h^2 - 3x^2}{h} = \frac{h\,(6x + 3h)}{h}$$
   $= 6x + 3h \;\to\; 6x.$
   That is 3 times the derivative of x^2: the constant just rides along.

2. $$\frac{(x+h)^2 + (x+h) - x^2 - x}{h} = \frac{2xh + h^2 + h}{h} = 2x + h + 1 \;\to\; 2x + 1.$$
   That is the derivative of x^2 plus the derivative of x: the sum splits.

3. At 4 hours the concentration is **falling**, at 1.5 mg/L per hour. If that pace held for one more hour, it would drop by about 1.5 mg/L.

Hold on to answers 1 and 2

They are two of the rules in the next section.

Derivative Rules

The Power Rule

Power rule

$$\frac{d}{dx}\, x^n = n\, x^{n-1} \qquad \text{for any number } n$$

Examples

function		as a power		derivative		simplified
x^3			$\longrightarrow$	$3x^2$		
x^{10}			$\longrightarrow$	$10x^9$		
$\sqrt{x}$	$=$	$x^{1/2}$	$\longrightarrow$	$\frac{1}{2}x^{-1/2}$	$=$	$\dfrac{1}{2\sqrt{x}}$
$\dfrac{1}{x^2}$	$=$	x^{-2}	$\longrightarrow$	$-2x^{-3}$	$=$	$-\dfrac{2}{x^3}$

Watch out

Rewrite roots and fractions as powers **first**: a root is a fractional power, a fraction a negative power.

Constant Multiple and Sum Rules

Constant multiple rule

$$\frac{d}{dx}\big[c\,f(x)\big] = c\,f'(x)$$

Constants ride along.

Sum rule

$$\frac{d}{dx}\big[f(x) \pm g(x)\big] = f'(x) \pm g'(x)$$

Sums and differences split.

Worked: a polynomial, term by term

$$\begin{aligned}\frac{d}{dx}\left(4x^3 - 2x + 7\right) &= 4 \cdot 3x^2 \;-\; 2 \cdot 1 \;+\; 0 \\ &= 12x^2 - 2\end{aligned}$$

Power rule on each term, constants carried, and the lone 7 (a constant) differentiates to 0.

Common Derivatives

Table

$f(x)$	$f'(x)$
c	0
x^n	$n\,x^{n-1}$
e^x	e^x
$\ln x$	$\dfrac{1}{x} \quad (x > 0)$
a^x	$a^x \ln a$
$\log_a x$	$\dfrac{1}{x \ln a} \quad (x > 0)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$

The Product Rule

Product rule

$$\frac{d}{dx}\big[f(x)\,g(x)\big] = f'(x)\,g(x) + f(x)\,g'(x)$$

In words: (first)′ × second + first × (second)′.

Worked: $x^2 e^x$

	$f = x^2, \quad g = e^x$	name the first and the second
	$f' = 2x, \quad g' = e^x$	differentiate each one
$\frac{d}{dx}\big[x^2 e^x\big]$	$= 2x \cdot e^x + x^2 \cdot e^x$	assemble $f'g + fg'$
	$= x\,e^x\,(2 + x)$	factor out $x\,e^x$

Notebook `U3-3_Derivatives-2_Rules`, §3

The Quotient Rule

Quotient rule

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)\,g(x) - f(x)\,g'(x)}{g(x)^2}$$

In words: "low d-high minus high d-low, over low squared" (low = bottom, high = top, d = derivative of).

Worked: $x/(x+1)$

$$\frac{d}{dx}\left[\frac{x}{x+1}\right]$$

$f = x, \quad g = x+1$ — name the top and the bottom

$f' = 1, \quad g' = 1$ — differentiate each one

$= \dfrac{1\cdot(x+1) - x\cdot 1}{(x+1)^2}$ — assemble $\dfrac{f'g - fg'}{g^2}$

$= \dfrac{x+1-x}{(x+1)^2} = \dfrac{1}{(x+1)^2}$ — simplify the top

Exercise: Rules

Try it: differentiate

1. $5x^4 - 3x^2 + 8$
2. $2\sqrt{x} + \dfrac{1}{x}$
3. $3e^x - 4\ln x$
4. $x^3 \ln x$
5. $\dfrac{e^x}{x}$
6. $(x+1)^2$ (expand it first)

Exercise: Solution

Answers

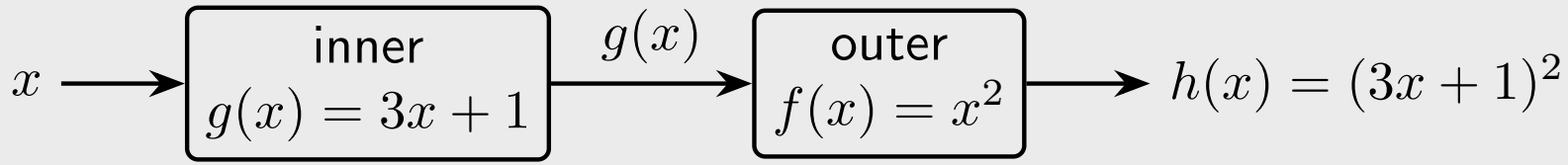
1. Term by term: $5 \cdot 4x^3 - 3 \cdot 2x + 0 = 20x^3 - 6x$.
2. As powers, $2x^{1/2} + x^{-1}$, so $2 \cdot \frac{1}{2}x^{-1/2} - x^{-2} = \dfrac{1}{\sqrt{x}} - \dfrac{1}{x^2}$.
3. From the table: $3e^x - 4 \cdot \dfrac{1}{x} = 3e^x - \dfrac{4}{x}$.
4. Product, with $f = x^3$, $g = \ln x$: $\quad 3x^2 \ln x + x^3 \cdot \dfrac{1}{x} = 3x^2 \ln x + x^2$.
5. Quotient, with top e^x, bottom x: $\quad \dfrac{e^x \cdot x - e^x \cdot 1}{x^2} = \dfrac{e^x(x-1)}{x^2}$.
6. $(x+1)^2 = x^2 + 2x + 1$, so the derivative is $2x + 2$. Expanding would not work for $(x+1)^{50}$: the chain rule, next, does it without expanding.

The Chain Rule

Functions Inside Functions

Composition, from Unit 1-3

$h(x) = (f \circ g)(x) = f\big(g(x)\big)$: the **inner** function g acts first, and the **outer** function f acts on its output.



Reading a composition backward

function h	inner g	outer f
$e^{-0.2t}$	$-0.2t$	e^t
$\sqrt{x^2 + 1}$	$x^2 + 1$	$\sqrt{x}$
$(5 - 2w)^2$	$5 - 2w$	w^2

Why we care

A model's error is a square of (actual minus predicted): the last row. Unit 3-2 differentiates exactly this.

The Chain Rule

Chain rule

$$h(x) = f\big(g(x)\big) \quad \Longrightarrow \quad h'(x) = f'\big(g(x)\big) \cdot g'(x)$$

In words: the derivative of the outer, at the inner, times the derivative of the inner.

Worked: $h(x) = (3x+1)^2$

	$g(x) = 3x+1, \quad f(x) = x^2$	name the inner and the outer
	$g'(x) = 3, \quad f'(x) = 2x$	differentiate each one
	$f'\big(g(x)\big) = 2(3x+1)$	outer's derivative, at the inner
$h'(x)$	$= 2(3x+1) \cdot 3 = 18x+6$	times $g'(x)$

Notebook `U3-3_Derivatives-2_Rules`, §4

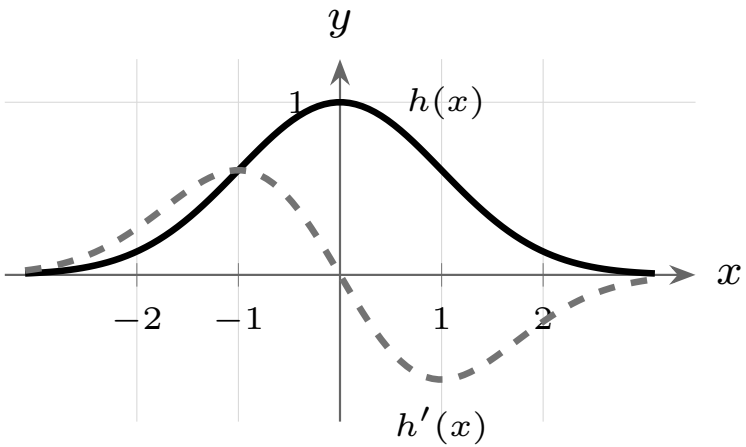
The Chain Rule: the Bell Curve

Worked: $h(x) = e^{-x^2/2}$, the shape of the normal distribution

$$g(x) = -\tfrac{1}{2}x^2, \quad f(x) = e^x$$ name the inner and the outer

$$g'(x) = -x, \quad f'(x) = e^x$$ differentiate each one

$$h'(x) = e^{-x^2/2} \cdot (-x) = -x\, e^{-x^2/2}$$ assemble $f'\big(g(x)\big) \cdot g'(x)$



What it says

- $h' > 0$ left of 0 (rising), $h' < 0$ right of 0 (falling), and $h'(0) = 0$ at the peak.
- The bell is steepest at $x = \pm 1$: one standard deviation from the center.

Exercise: Chain Rule

Try it: differentiate

1. $(2x - 5)^3$
2. e^{3x}
3. $\sqrt{x^2 + 1}$

Exercise: Solution

Answers

1. Inner $g = 2x - 5$ ($g' = 2$), outer $f = x^3$ ($f' = 3x^2$):
   $3(2x-5)^2 \cdot 2 = 6(2x-5)^2$.
2. Inner $g = 3x$ ($g' = 3$), outer $f = e^x$ ($f' = e^x$): $\quad e^{3x} \cdot 3 = 3e^{3x}$.
3. Inner $g = x^2 + 1$ ($g' = 2x$), outer $f = x^{1/2}$ ($f' = \frac{1}{2}x^{-1/2}$):

$$\tfrac{1}{2}(x^2+1)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2+1}}.$$

Partial Derivatives

Functions of Two Variables, from Unit 1-3

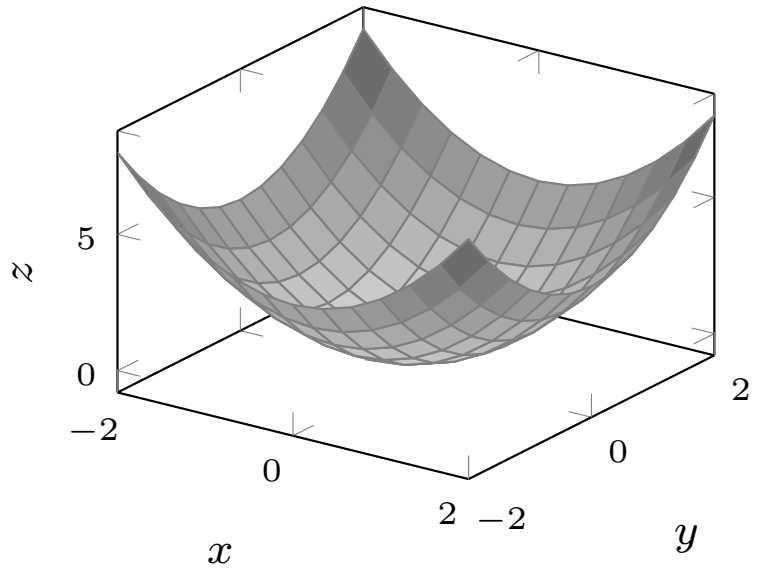
Two inputs, one output

$f(x, y)$ takes a **pair** of numbers and returns **one**. Its graph is a **surface**: each point (x, y) on the floor gets a height $z = f(x, y)$.

Examples

body mass index	$B(w, h) = w/h^2$
drug concentration	$C(D, V) = D/V$
a bowl	$f(x, y) = x^2 + y^2$
a model's loss	$L(m, b)$

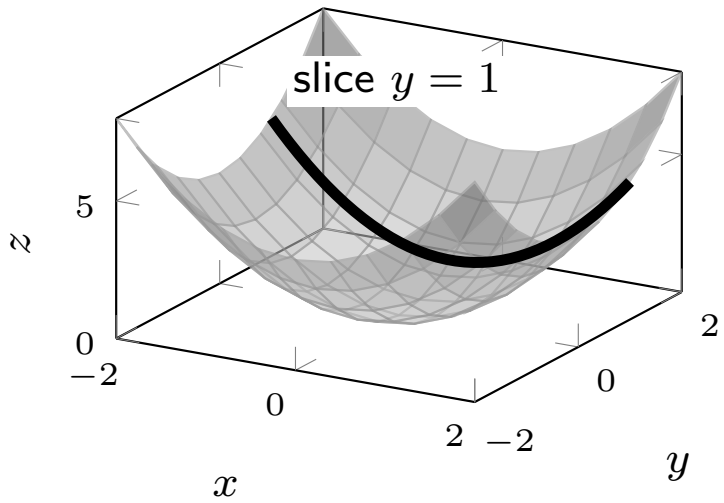
Bowl: $z = x^2 + y^2$

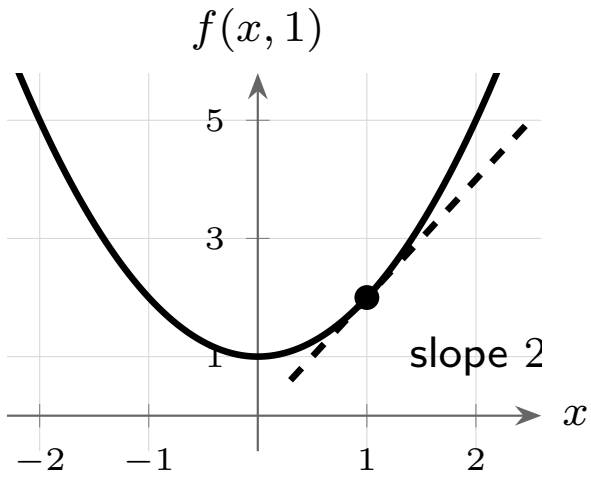


One lowest point, at $(0, 0)$.

Notebook `U3-6_Multivariable-1_Surface3D`, §1 and §2

Freeze One Variable





Dashed: the tangent at $x = 1$.

The idea, for $f(x, y) = x^2 + y^2$

- Hold $y = 1$ fixed: $f(x, 1) = x^2 + 1$ is a function of x **alone**, the slice of the bowl.
- Differentiate it with the rules from earlier: slope $2x$, so slope 2 at $x = 1$.
- That is the **partial derivative of** f **with respect to** x at $(1, 1)$.

Notebook `U3-6_Multivariable-2_PartialDerivatives`, §1 and §3

The Definition

Partial derivatives as difference quotients

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h,\, y) - f(x,y)}{h} \qquad \text{only } x \text{ moves}$$

$$\frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x,\, y+h) - f(x,y)}{h} \qquad \text{only } y \text{ moves}$$

The same difference quotient as before, with the other variable frozen.

Notation, and how to compute one

- ∂ is a curly d, read "partial": $\frac{\partial f}{\partial x}$ is "partial f, partial x". Shorthand: f_x and f_y.
- The curly d is a warning: other variables exist and are being held fixed.
- **In practice**: treat every other variable as a **constant**, then use the rules from earlier in this lecture.

What a Partial Means: BMI

$$B(w, h) = \frac{w}{h^2} = w\,h^{-2}$$

- Freeze h: B is w times the constant h^{-2}, so $\dfrac{\partial B}{\partial w} = \dfrac{1}{h^2}$.
- Freeze w: power rule on h^{-2}, so $\dfrac{\partial B}{\partial h} = w \cdot (-2)h^{-3} = -\dfrac{2w}{h^3}$.

For a patient with $w = 70$ kg and $h = 1.75$ m

- $\dfrac{\partial B}{\partial w} = \dfrac{1}{1.75^2} \approx 0.33$ BMI points per kg: one more kg raises BMI by about a **third** of a point.
- $\dfrac{\partial B}{\partial h} = -\dfrac{2 \cdot 70}{1.75^3} \approx -26$ BMI points per m, about -0.26 per cm: one more cm of height, at the same weight, lowers BMI by about a **quarter** of a point.

Units: a partial is in output units **per** unit of the input that moved.

Review: Limits and Derivatives

What we covered: from secant to tangent

1. The **average rate of change** from x_1 to x_2 is $\frac{f(x_2)-f(x_1)}{x_2-x_1}$: the slope of a **secant line**. Its units are output per input.
2. The **difference quotient** $\frac{f(x+h)-f(x)}{h}$ is the same slope, from x to $x+h$.
3. Setting $h=0$ gives $0/0$. Instead take a **limit**: let h get as close to 0 as we like, without reaching it. Simplify first, then let h shrink.
4. The **derivative** $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ is the slope of the **tangent line**: the **instantaneous** rate of change. Also written $\frac{df}{dx}$ and $\frac{d}{dx}f(x)$.
5. The sign of f' says rising (>0), falling (<0), or flat ($=0$).

Review: Limits and Derivatives

What we covered: the rules

6. From the definition: $c \to 0$, $x \to 1$, $mx + b \to m$, $x^2 \to 2x$, $\sin x \to \cos x$, $\frac{1}{x} \to -\frac{1}{x^2}$.
7. **Power rule**: $x^n \to n\,x^{n-1}$. Rewrite roots and fractions as powers first.
8. Constants **ride along**; sums and differences **split**.
9. **Product rule**: $(fg)' = f'g + fg'$. **Quotient rule**: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$.
10. The table: $e^x \to e^x$, $\ln x \to \frac{1}{x}$, $a^x \to a^x \ln a$, $\log_a x \to \frac{1}{x \ln a}$, $\sin x \to \cos x$, $\cos x \to -\sin x$.
11. **Chain rule**: for a composition $h(x) = f(g(x))$, $h'(x) = f'(g(x)) \cdot g'(x)$: the outer's derivative at the inner, times the inner's derivative.

Review: Limits and Derivatives

What we covered: two variables

12. A function $f(x, y)$ of two variables has a **surface** for a graph (Unit 1-3).
13. A **partial derivative** freezes the other variable and differentiates with the same rules: $\frac{\partial f}{\partial x}$ is the slope of the slice in the x direction, in output units per input unit.

Notebooks: `U3-2_Limits-1`; `U3-3_Derivatives-1`, `-2`; `U3-6_Multivariable-1`, `-2`.
Previous: Unit 2-5, Determinants and Inverses. **Next:** Unit 3-2, **Optimization and Linear Regression**.